

Customer Behavior Analysis for Alfido Tech

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Google Colab Notebook Link:

https://colab.research.google.com/drive/1Xg-_oPU4cKUO3WSWWU9xOE2i1DYU5u_-?usp=sharing

Executive Summary

This project analyzes customer behavior using an e-commerce dataset containing 250,000 customer records. Data cleaning and preprocessing were performed to prepare the dataset for analysis. Customer segmentation was conducted using RFM (Recency, Frequency, Monetary) metrics and K-Means Clustering to identify groups of customers with similar purchasing patterns.

The analysis revealed that a small group of high-value customers contributes significantly to revenue, while some customers exhibit churn behavior and require retention strategies. Product category analysis showed differences in revenue contribution across categories, and payment analysis indicated a strong preference for digital payment methods. Monthly revenue trends highlighted variations in purchasing activity over time.

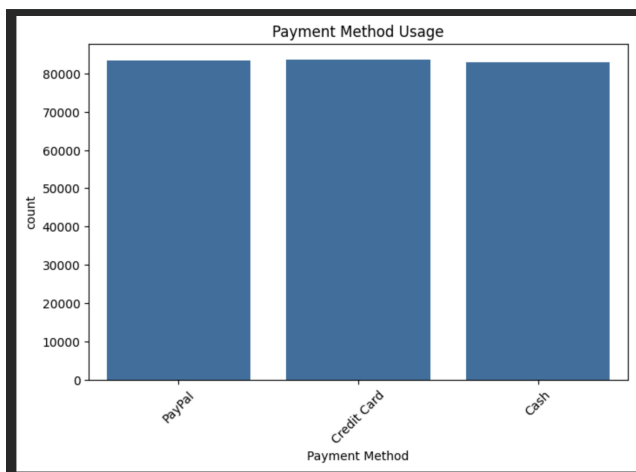
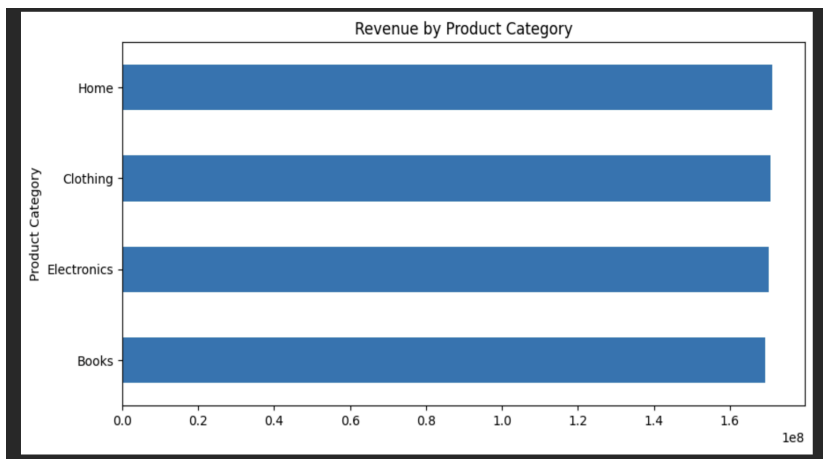
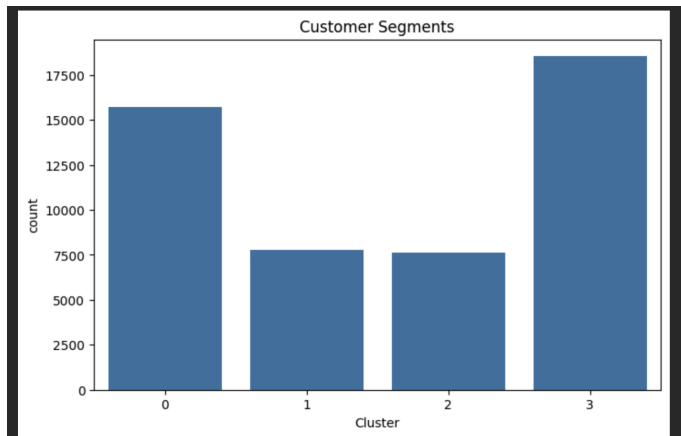
Key Findings

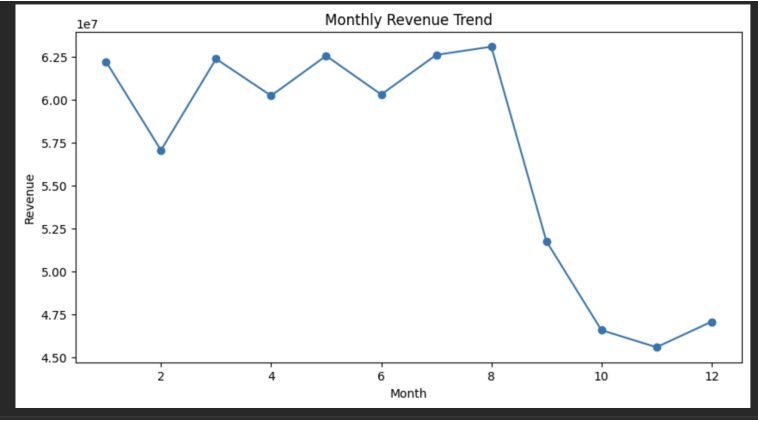
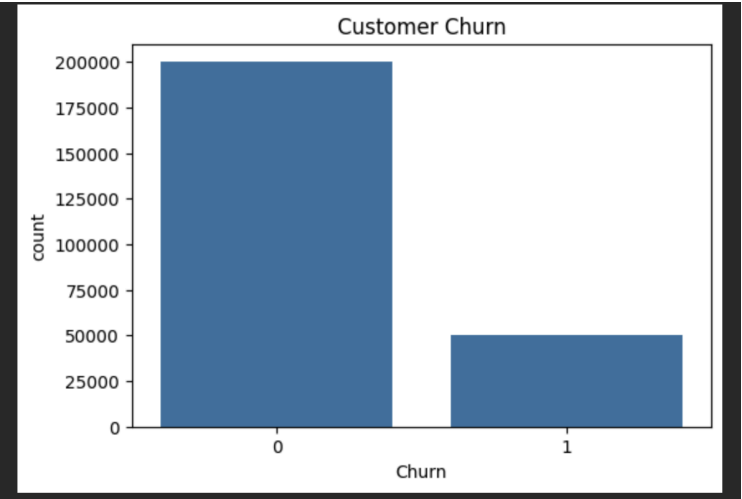
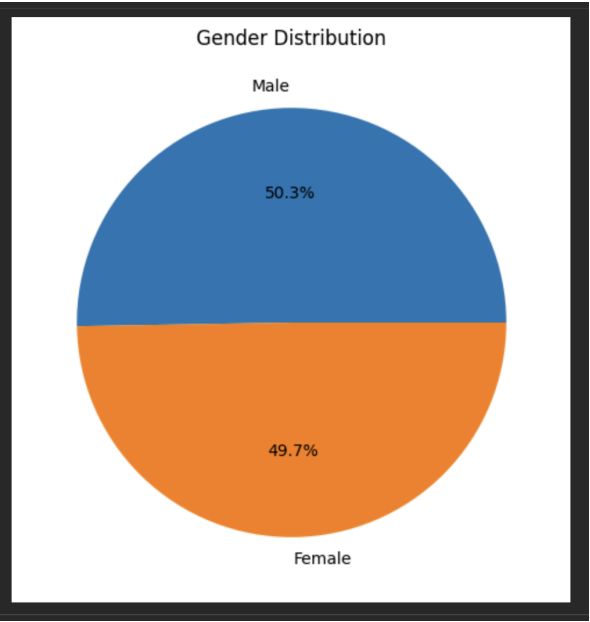
- High-value customers generate a major share of revenue.
- Customer churn is present and requires targeted engagement.
- Product categories contribute differently to sales performance.
- Digital payment methods are the most commonly used.
- Purchasing patterns vary across customer segments.

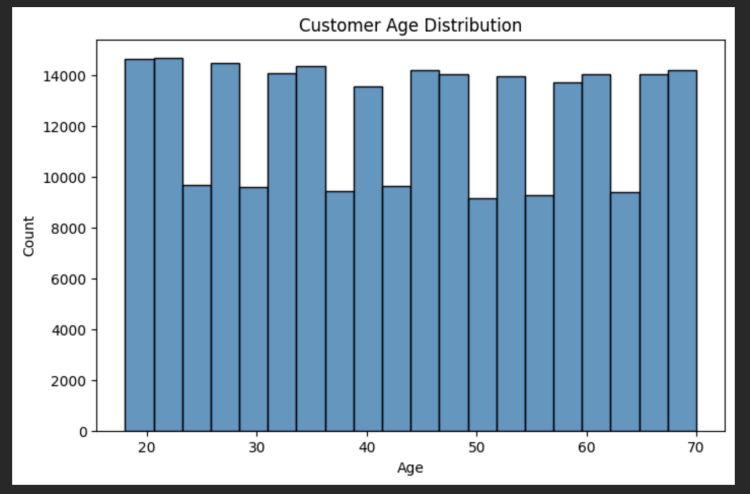
Recommendations

1. Implement loyalty programs for high-value customers.
2. Offer personalized promotions to at-risk customers.
3. Increase focus on top-performing product categories.
4. Use recommendation systems based on purchase history.
5. Strengthen customer retention through targeted marketing.

Visualizations







Conclusion

The analysis successfully identified customer segments, purchasing trends, and churn behavior. These insights can help Alfido Tech improve customer retention, customer satisfaction, and overall business performance.